

1) Larry came home very excited after a visit to his doctor. He announced proudly to his wife, “My doctor says my blood pressure is at the 90th percentile among men like me. That means I’m better off than about 90% of similar men.” How should his wife, who is a statistician, respond to Larry’s statement?

2) The distribution of heights of adult American men is approximately normal with mean 69 inches and standard deviation 2.5 inches. Draw a normal curve below and mark three standard deviations out on either side. Then, use the curve to answer the following questions.

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| (a) What percent of men are shorter than 64 inches? | (b) What percent of men are taller than 71.5 inches? |
| (c) What percent of men are taller than 66.5 inches? | (d) A height of 74 inches corresponds to what percentile of American male heights? |

3) The length of human pregnancies from conception to birth varies on an approximate normal distribution with mean 266 days and a standard deviation of 6 days. Draw a normal curve below and mark three standard deviations out on either side. Then, use the curve to answer the following questions.

a) How long are the longest 16% of pregnancies?

b) How short are the shortest 16% of pregnancies?

c) Between how many days does the middle 95% of all pregnancies fall?

d) Between how many days does the middle 99.7% of all pregnancies fall?

e) A pregnancy of 254 days corresponds to what percentile of pregnancies?

4) Three landmarks of baseball achievement are Ty Cobb's batting average of 0.420 in 1911, Ted Williams's 0.406 in 1941, and George Brett's 0.390 in 1980. These batting averages cannot be compared directly because the distribution of major league batting averages has changed over the years. The distributions are quite symmetric, except for outliers such as Cobb, Williams, and Brett. While the mean batting average has been held roughly constant by rule changes and the balance between hitting and pitching, the standard deviation has dropped over time. Here are the facts:

Decade	Mean	Standard Deviation
1910s	0.266	0.0371
1940s	0.267	0.0326
1970s	0.261	0.0317

Find the standardized scores for Cobb, Williams, and Brett. Who was the best hitter?

5) The scores of an introductory statistics class at the University of Wisconsin is normally distributed with a mean of 72 and standard deviation of 10.

(a) If a student has a z-score of 1.4, what score does she have?

(b) If a student has a z-score of -1.23, what score does he have?

6) A college statistics exam scores were normally distributed with $\mu = 150$ points and $\sigma = 20$ points. The z-scores for some students are below. Find the final exam raw score for each.

Chris 0.1

Janet 1.8

Rick -1.1

5) Make an accurate sketch of a Standard Normal Distribution and answer each question using a standard normal curve.

a) What is the area between $z = 0$ and $z = 1$?

b) What percent of the data is within one standard deviation of the mean?

c) What percent of the data is more than one standard deviation from the mean?

d) What is the area between $z = 0$ and $z = 2$?

e) What is the area under the curve between $z = -1$ and $z = 2$?